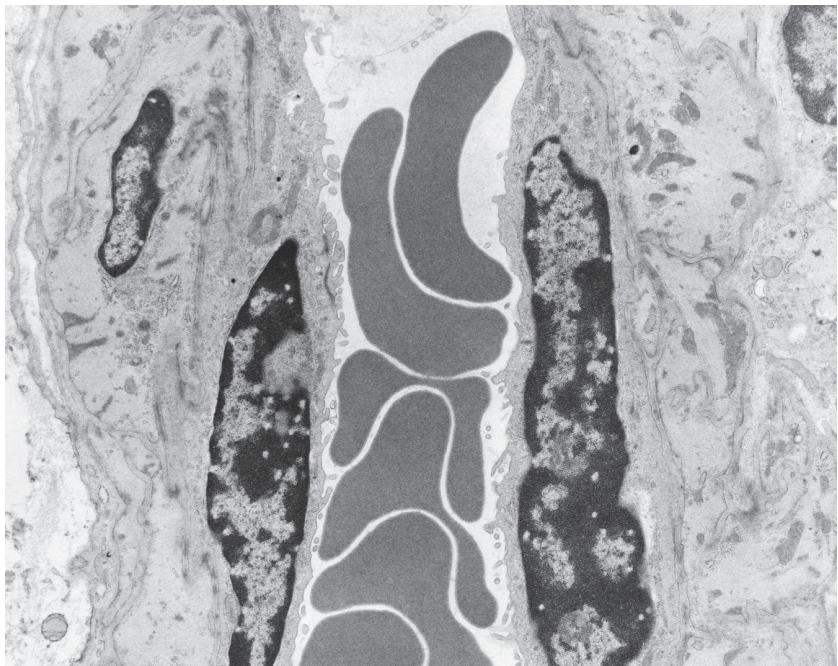


- 3 Fig. 3.1 is a transmission electron micrograph showing red blood cells in a capillary of a healthy adult.



magnification = $\times 4000$

Fig. 3.1

- (a) (i) Explain how the cells in the capillary shown in Fig. 3.1 can be identified as red blood cells.

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- (ii) Capillaries are surrounded by tissue fluid.

Outline the ways in which the composition of tissue fluid differs from blood.

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- (b) Scientists compared features of the physiology of three groups of people who live at different altitudes. These people were selected from populations that have lived at these altitudes for many thousands of years.

The scientists took blood samples from people in each location and measured:

- the concentration of haemoglobin in the blood
- the percentage saturation of haemoglobin with oxygen in blood leaving the lungs
- the oxygen concentration in the blood leaving the lungs.

The results are shown in Table 3.1.

Table 3.1

altitude / m	location of group of people	mean haemoglobin concentration in blood / g 100 cm ⁻³	mean percentage saturation of haemoglobin with oxygen in blood leaving the lungs	mean oxygen concentration in blood leaving the lungs / cm ³ 100 cm ⁻³
100 (near sea level)	Peru	15.3	97.0	21.1
3500	East Africa mountains	15.6	95.5	21.1
3750 – 4000	Andes mountains	19.1	92.0	22.0

- (i) Describe the changes in mean percentage saturation of haemoglobin with oxygen and the mean haemoglobin concentration in blood as altitude increases.

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- (ii) With reference to Table 3.1, suggest how the people living at high altitude can have an oxygen concentration in blood leaving the lungs similar to that of the people living at sea level.

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[Total: 11]